



Technical Service Bulletin: tsb101149

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Aftercooler Core Improvement, and Air Intake Manifold and Aftercooler Cover Rework

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Note : This service/parts topic supersedes 00T10-4; replace 00T10-4 with this topic.

This service/parts topic introduces changes to the aftercooler, air intake manifold and aftercooler cover for the QSK60 engine. The topic can be split into four sections:

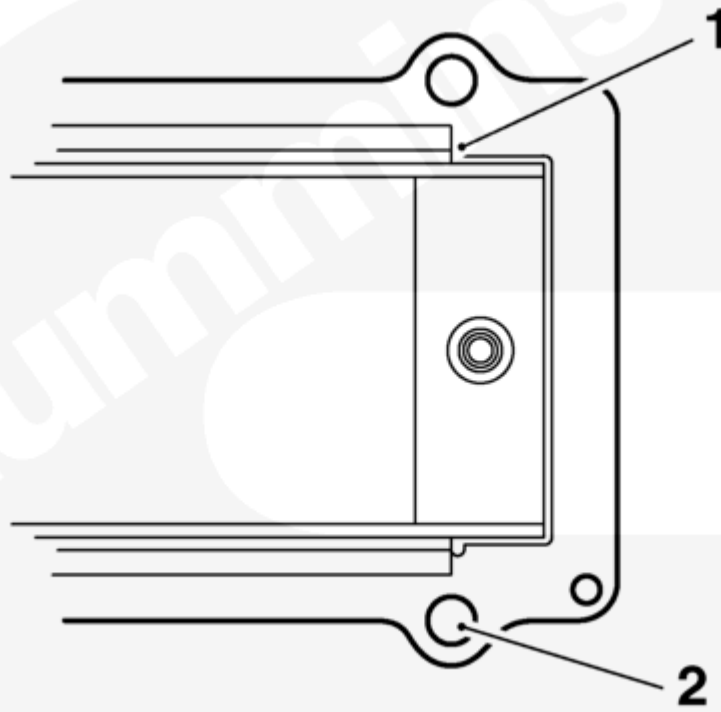
1. New parts introduction.
2. Compatibility matrix.
3. Rework/Machine procedure.
4. Old aftercooler core reuse guidelines and salvage part.

A new aftercooler core has been introduced to address a cracking issue with the old aftercooler core flange. The change in the aftercooler core design has also led to changes in the intake manifold and the aftercooler cover to accommodate the new aftercooler core.

The following changes have been made to the aftercooler core to improve strength and add durability:

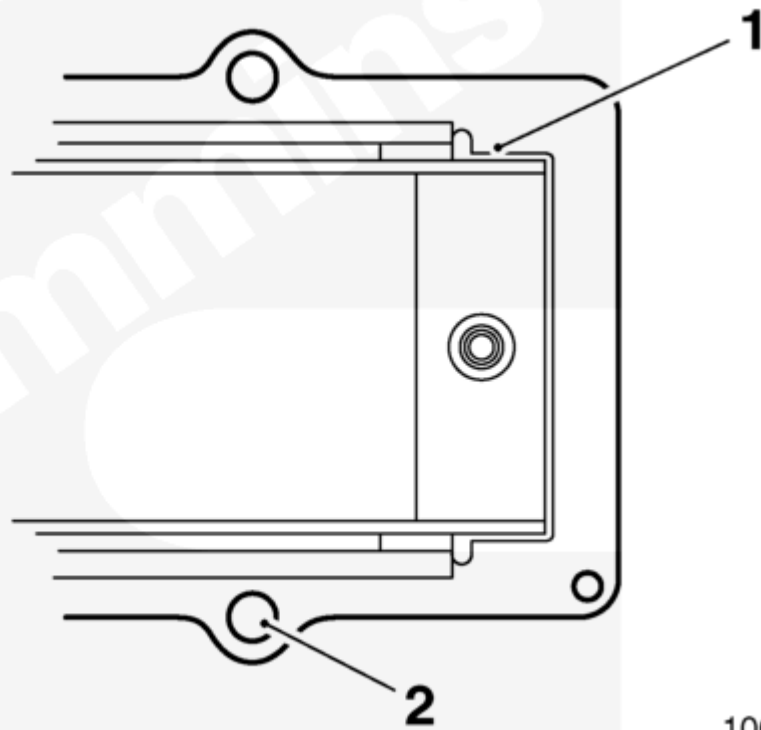
1. Relocation of the aftercooler core capscrew mounting holes inboard of the flange-forming notch, this is similar to the QSK45 and KV38 and KV50 products. Figure 1 shows the old design; Figure 2 shows the new design.
2. Relocation of the flange welds. See Figure 3.
3. 30% increase in flange thickness, the old flange was 3.42 mm [0.135 in] the new flange is 4.55 mm [0.179 in] thick. Figure 4.

The new aftercooler core went into production at the Daventry Engine Plant with engine serial number 33150882



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Fig. 1, Flange Forming Notch (1), Old Capscrew Location (2).



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Fig 2, Flange Forming Notch (1), New Capscrew Location (2).

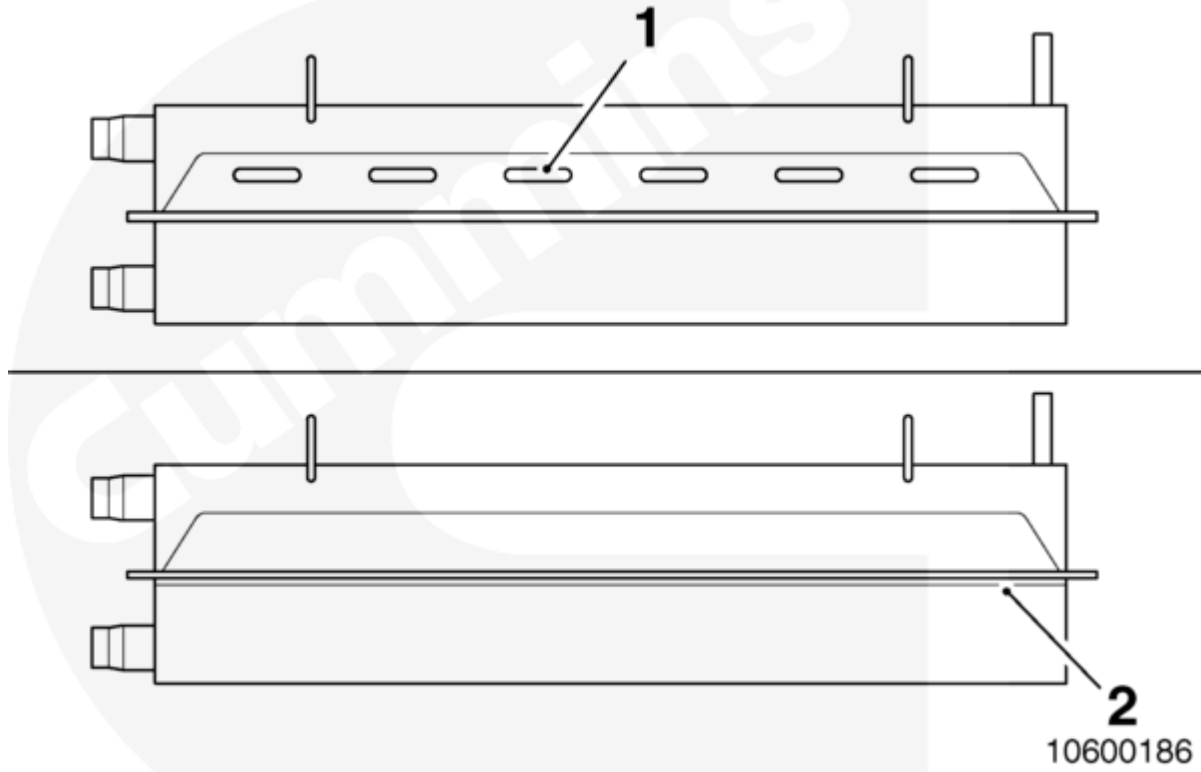


Fig. 3, New Weld Location in middle of flange (1), Old Weld Location on bottom of flange (2).

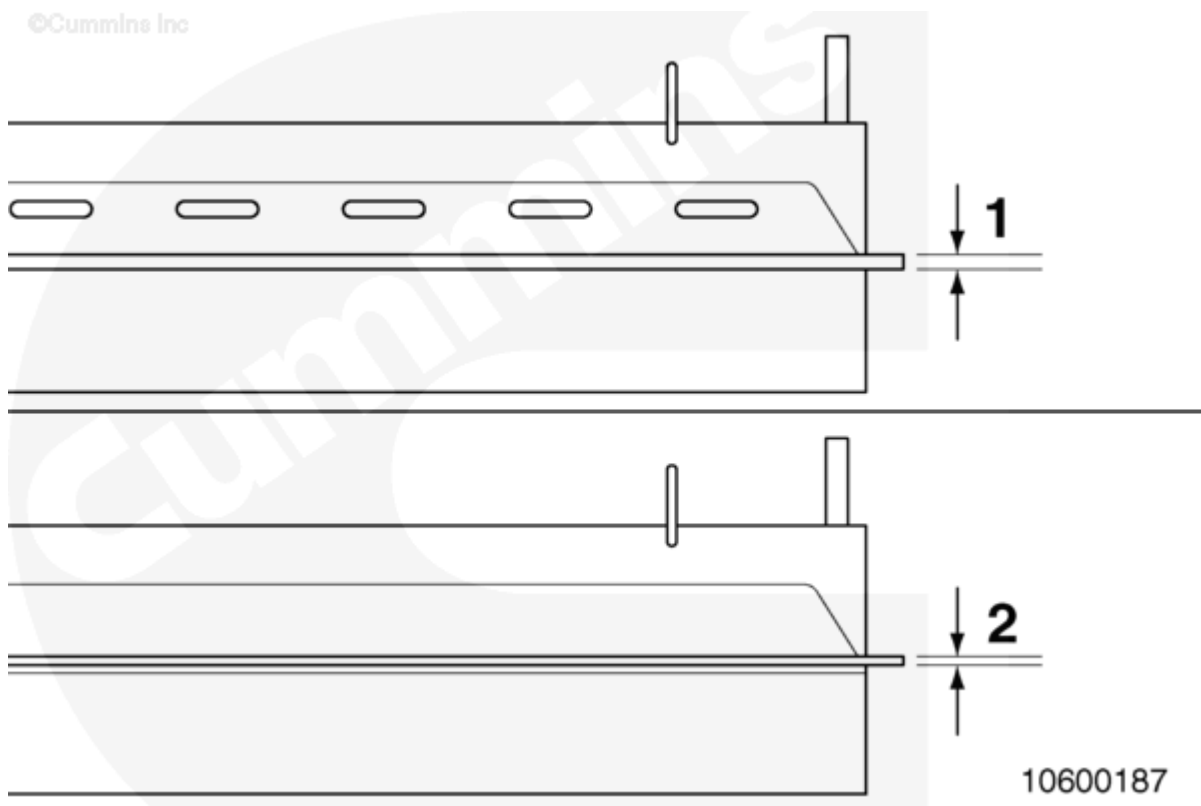


Fig. 4, New design 4.55 mm [0.179 in] (1), Old design 3.42 mm [0.135 in] (2).

The air intake manifold and aftercooler cover have been modified to accept the new aftercooler core. The new intake manifold will **not** accept the old aftercooler core. The old manifold and cover will **not** accept the new aftercooler core without rework/machining, the rework/machining procedure is given below. If a new cover is fitted to an old intake manifold then a new longer roll pin will be required to align the cover.

Parts Information				
Description	Old Part Number	Service Parts Available	New Part Number	Service Parts Available
Manifold, Air Intake	4017703	No	4080571	Yes
Manifold, Air Intake	4017704	No	4080572	Yes
Cover, Aftercooler	4001329	No	4080569	Yes
Cover, Aftercooler	4001330	No	4080570	Yes
Gasket, Aftercooler Cover	4000713	Yes	4080573	Yes
Core, Aftercooler	4001326	Yes	4080568	Yes
Pin, Roll	N/A	N/A	4086080	Yes
Plate, Clamping	N/A	N/A	4066244	Yes

Compatibility Matrix						
	Old Aftercooler Core	New Aftercooler Core	Old Air Intake Manifold	New Air Intake Manifold	Old Aftercooler Cover	New Aftercooler Cover
Old Aftercooler Core			Y	N	Y	Y
New Aftercooler Core			M	Y	M	Y
Old Air Intake Manifold	Y	M			Y	D
New Air Intake Manifold	N	Y			N	Y
Old Aftercooler Cover	Y	M	Y	N		

Compatibility Matrix						
	Old Aftercooler Core	New Aftercooler Core	Old Air Intake Manifold	New Air Intake Manifold	Old Aftercooler Cover	New Aftercooler Cover
New Aftercooler Cover	Y	Y	D	Y		
Clamping Plate	Y	N	Y	N	Y	Y
Chart Key Y = Yes, will fit with no modifications required. N= No, will not fit. M = Will fit after modification, see machine procedure for details of the modification. D = Longer dowel pins are required to align the aftercooler cover and the air intake manifold. Note : This is a product improvement and is not subject to campaign.						

Last Modified: 15-May-2002
